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COM3502-4502-6502 SPEECH PROCESSING

Lecture 17
Adaptive Filters

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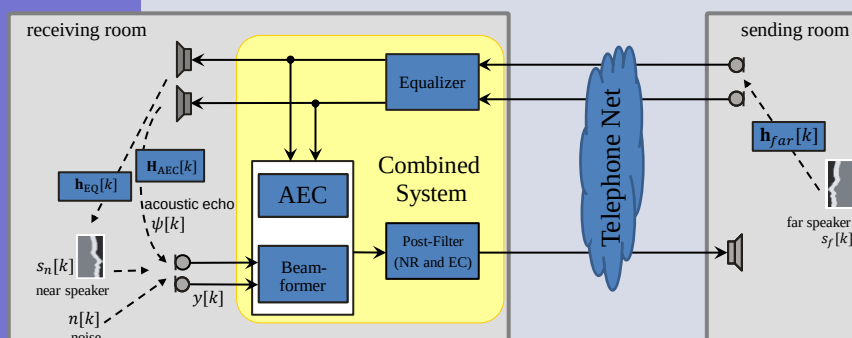
Examples (Hands-free systems)

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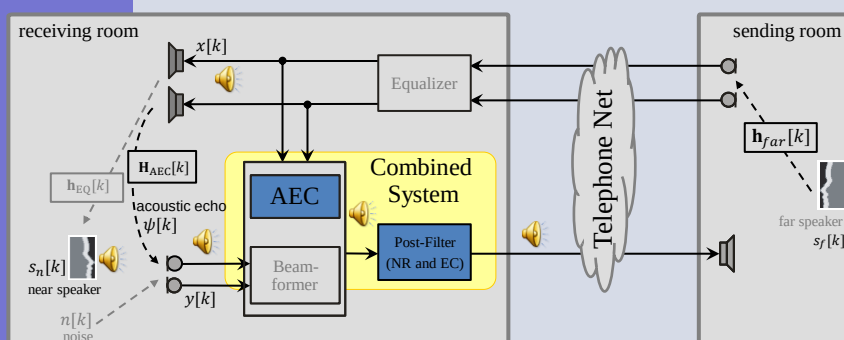
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Model of Hands-Free System



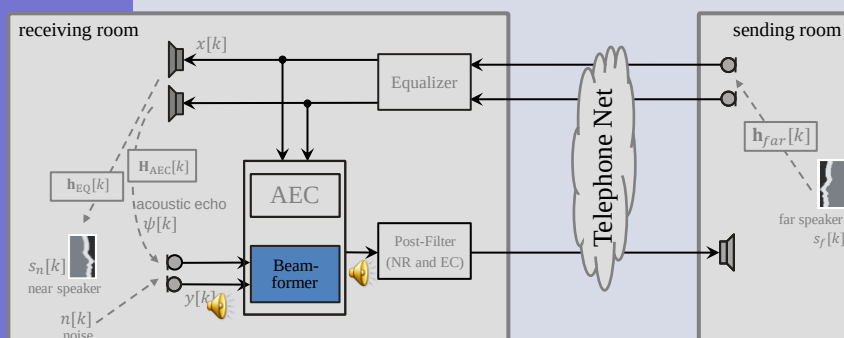
- All three problems (noise reduction (NR), acoustic echo cancellation (AEC), dereverberation / listening-room compensation (LRC)) have to be solved in combination.
- Synergies can be exploited when algorithms are designed jointly.

Demot: Acoustic Echo Cancellation



- The far-end speaker would hear his/her own signal $s_f[k]$ delayed by the round-trip delay of the electro acoustic overall system
 - This severely hinders humans ability to express oneself (speech planning)

Demo: Noise Reduction (MC)



- Noise reduction algorithms may rely on only one microphone (single-channel, SC) or multiple-microphones (multi-channel, MC) to exploit spatial information

Applications for Adaptive Filters

- Areas of application:
 - Speech coding (e.g. GSM, UMTS)
 - Speech signal improvement (Hands-free systems, Hearing aids, etc.)
 - Equalization (Transmitters, Radar, Loudspeaker)
 - Anti-sound systems (Vehicle and Airplane Technology)
 - Multi-channel processing (Beamforming, Submarine detection)
 - Medical technology (Measurement of the heart beats, etc.)
 - Image signal processing (Compensation of image interferences, image analysis)
 - Antenna fields and array signal processing (e.g. noise reduction)

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FIR Filter (Constant Coefficients)

Transversal Filter

$$\mathbf{x}[k] = [x[k], x[k-1], x[k-2], \dots, x[k-L_{\text{AEC}}+1]]^T$$
$$\mathbf{w}_{\text{AEC}} = [w_{\text{AEC},0}, w_{\text{AEC},1}, w_{\text{AEC},2}, \dots, w_{\text{AEC},L_{\text{AEC}}-1}]^T$$
$$\hat{\psi}[k] = \mathbf{w}_{\text{AEC}}[k]^H \mathbf{x}[k] = \mathbf{x}^H[k] \mathbf{w}_{\text{AEC}}[k]$$

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FIR Filter (Time-Varying)

$$\mathbf{x}[k] = [x[k], x[k - 1], x[k - 2], \dots, x[k - L_{AEC} + 1]]^T$$
$$\mathbf{w}_{AEC}[k] = [w_{AEC,0}[k], w_{AEC,1}[k], w_{AEC,2}[k], \dots, w_{AEC,L_{AEC}-1}[k]]^T$$
$$\hat{y}[k] = \mathbf{w}_{AEC}[k]^H \mathbf{x}[k] = \mathbf{x}^H[k] \mathbf{w}_{AEC}[k]$$

time-varying coefficients

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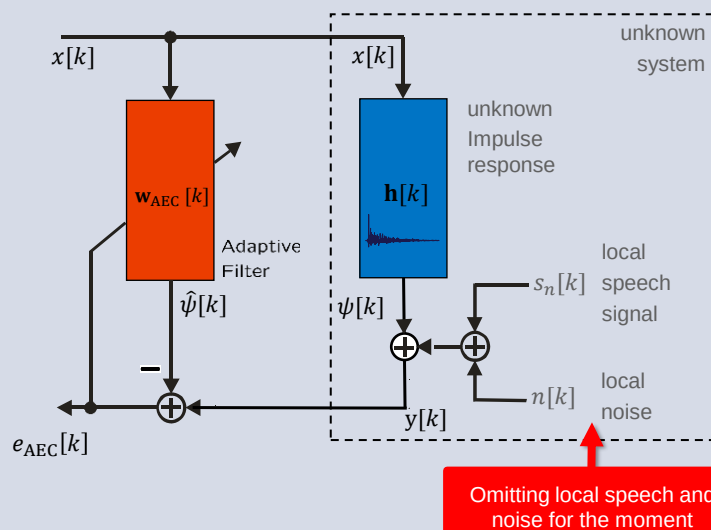
Design of Adaptive Filters

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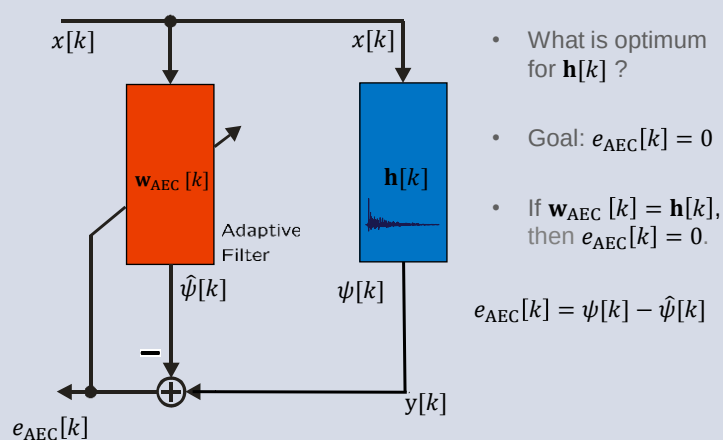
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First example of an Adaptive Filter – The AEC



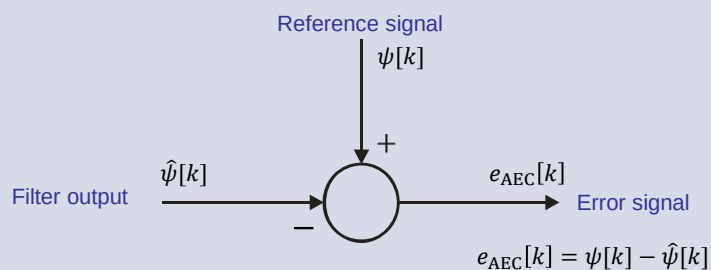
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First example of an Adaptive Filter – The AEC



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Cost Functions for Adaptive Filters



- What signals are available or can be estimated.
- How to minimize such a cost function $e_{\text{AEC}}[k]$?

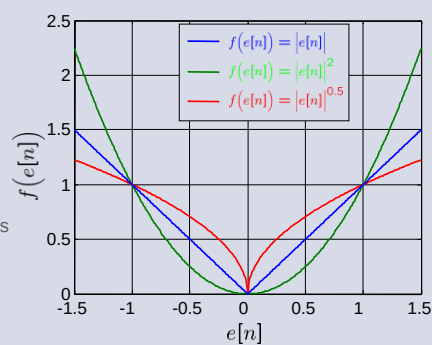
Possible Cost Functions

Examples:

- $f(e[k]) = |e[k]|$
- $f(e[k]) = |e[k]|^2$
- generally: $f(e[k]) = |e[k]|^\alpha$
 - $\alpha > 1$ amplification of large errors
 - $\alpha < 1$ weakens large errors

- Mean squared error:
 $f(e[k]) = E\{|e[k]|^2\}$

$e[k]$: stochastic process

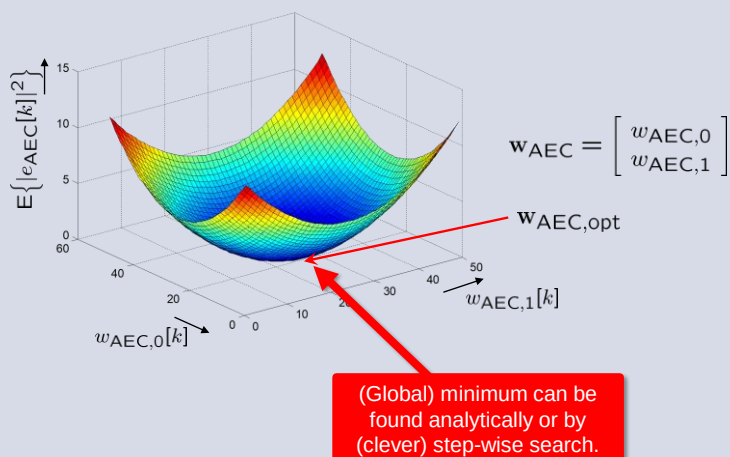


- How to calculate minimum of (quadratic) function?

$$E\{|e[k]|^2\} \stackrel{!}{=} \min$$

Error Performance Surface

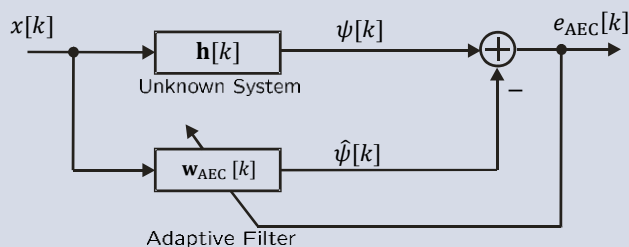
Preferred: Square Cost Function $\Rightarrow$ Mean square Error



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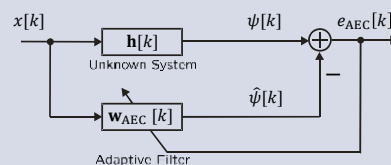
Objective Function

- Objective function:
 - Minimization of the mean square error $E\{|e_{AEC}[k]|^2\} \stackrel{!}{=} \min$
- Assumption:
 - Stationary **real** stochastic processes



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Wiener Filter in Time-Domain - Derivation



- Output of the adaptive Filter:

$$\hat{\psi}[k] = \sum_{i=0}^{L_{AEC}-1} w_{AEC,i}[k] x[k-i] = \mathbf{w}_{AEC}^T[k] \mathbf{x}[k] = \mathbf{x}^T[k] \mathbf{w}_{AEC}[k]$$

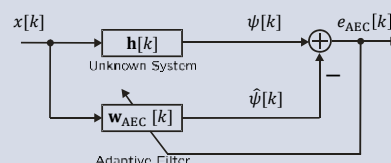
- Error:

$$\begin{aligned} e_{AEC}[k] &= \psi[k] - \hat{\psi}[k] \\ &= \psi[k] - \mathbf{w}_{AEC}^T[k] \mathbf{x}[k] = \psi[k] - \mathbf{x}^T[k] \mathbf{w}_{AEC}[k] \end{aligned}$$

- Mean square error (MSE):

$$E\{|e_{AEC}[k]|^2\} = E\{[\psi[k] - \hat{\psi}[k]]^2\} = E\{[\psi[k] - \mathbf{w}_{AEC}^T[k] \mathbf{x}[k]]^2\}$$

Wiener Filter in Time-Domain - Derivation

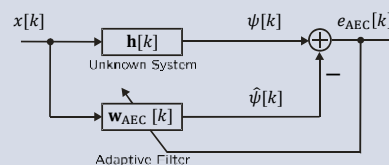


- Squared error:

$$\begin{aligned} |e_{AEC}[k]|^2 &= (\psi[k] - \hat{\psi}[k])^2 \\ &= (\psi[k] - \mathbf{w}_{AEC}^T[k] \mathbf{x}[k])^2 \\ &= (\psi[k] - \mathbf{w}_{AEC}^T[k] \mathbf{x}[k]) (\psi[k] - \mathbf{w}_{AEC}^T[k] \mathbf{x}[k])^T \\ &= (\psi[k] - \mathbf{w}_{AEC}^T[k] \mathbf{x}[k]) (\psi[k] - \mathbf{x}^T[k] \mathbf{w}_{AEC}[k]) \\ &= \psi^2[k] - \mathbf{w}_{AEC}^T[k] \mathbf{x}[k] \psi[k] - \psi[k] \mathbf{x}^T[k] \mathbf{w}_{AEC}[k] \\ &\quad + \mathbf{w}_{AEC}^T[k] \mathbf{x}[k] \mathbf{x}^T[k] \mathbf{w}_{AEC}[k] \end{aligned}$$

$$\begin{aligned} |\mathbf{A}|^2 &= \mathbf{A} \mathbf{A}^T = \mathbf{A}^T \mathbf{A} \\ (\mathbf{A} + \mathbf{B})^T &= \mathbf{A}^T + \mathbf{B}^T \\ (\mathbf{A} \mathbf{B})^T &= \mathbf{B}^T \mathbf{A}^T \end{aligned}$$

Wiener Filter in Time-Domain - Derivation



- Finding the **mean** of the **minimum** of a quadratic (positive) quantity:

– Derive towards filter coefficients and set to zero

$$\frac{\partial E\{|e_{AEC}[k]|^2\}}{\partial \mathbf{w}_{AEC}[k]} \stackrel{!}{=} 0$$

- Minimum Mean Square Error** Approach

Wiener Filter in Time-Domain - Derivation

- Mean square error:

$$E\{|e_{AEC}[k]|^2\} = E\{\psi^2[k] - \mathbf{w}_{AEC}^T[k] \mathbf{x}[k] \psi[k] - \psi[k] \mathbf{x}^T[k] \mathbf{w}_{AEC}[k] - \mathbf{w}_{AEC}^T[k] \mathbf{x}[k] \mathbf{x}^T[k] \mathbf{w}_{AEC}[k]\}$$

- Derivative:

$$\begin{aligned} \frac{\partial E\{|e_{AEC}[k]|^2\}}{\partial \mathbf{w}_{AEC}^T[k]} &= E\left\{-\mathbf{x}[k] \psi[k] - \left(\psi[k] \mathbf{x}^T[k]\right)^T \right. \\ &\quad \left. + \mathbf{x}[k] \mathbf{x}^T[k] \mathbf{w}_{AEC}[k] + \underbrace{\left(\mathbf{w}_{AEC}^T[k] \mathbf{x}[k] \mathbf{x}^T[k]\right)^T}_{\mathbf{x}[k] \mathbf{x}^T[k] \mathbf{w}_{AEC}[k]}\right\} \\ &= -2 E\{\mathbf{x}[k] \psi[k]\} + 2 E\{\mathbf{x}[k] \mathbf{x}^T[k]\} \mathbf{w}_{AEC}[k] \\ &\stackrel{!}{=} 0 \end{aligned}$$

$$\underbrace{E\{\mathbf{x}[k] \psi[k]\}}_{\mathbf{r}_{x\psi}[k]} = \underbrace{E\{\mathbf{x}[k] \mathbf{x}^T[k]\}}_{\mathbf{R}_{xx}[k]} \mathbf{w}_{AEC}[k]$$

$$\mathbf{w}_{AEC,opt} = \mathbf{R}_{xx}^{-1} \mathbf{r}_{x\psi}$$

Wiener Filter in Time-Domain - Derivation

- Alternative approach (knowing optimum filter theory):
- Filter coefficients are optimal with **orthogonality**:

$$E\{x[k] e_{\text{AEC},\min}[k]\} = E\{x[k] [\psi[k] - \mathbf{w}_{\text{AEC},\text{opt}}^T x[k]]\} = 0$$
- Abbreviations:

$$\mathbf{R}_{xx} = E\{x[k] x^T[k]\} \quad (\text{Autocorrelation matrix})$$

$$\mathbf{r}_{x\psi} = E\{x[k] \psi[k]\} \quad (\text{Cross correlation vector})$$
- Solution (Wiener Equation / Normal Equation):

$$\mathbf{R}_{xx} \mathbf{w}_{\text{AEC},\text{opt}} = \mathbf{r}_{x\psi} \quad \boxed{\mathbf{w}_{\text{AEC},\text{opt}} = \mathbf{R}_{xx}^{-1} \mathbf{r}_{x\psi}}$$

(If inverse exists)

Problems and Goals

- Problems:
 - huge matrix $\mathbf{R}_{xx}$ has to be inverted, size of $\mathbf{R}_{xx}[k]$ is $L_{\text{AEC}} \times L_{\text{AEC}}$ thus, e.g. 1024×1024
 - $\mathbf{R}_{xx}[k] = E\{x[k] x^T[k]\}$ and $\mathbf{r}_{x\psi}[k] = E\{x[k] \psi[k]\}$ must be estimated
 - input signal $x[k]$ and echo signal $\psi[k]$ are non-stationary and time-varying
$$\begin{aligned} \hat{\mathbf{R}}_{xx}[k] &= x[k] x^T[k] \\ \hat{\mathbf{r}}_{x\psi}[k] &= x[k] \psi[k] \end{aligned} \quad \left. \vphantom{\begin{aligned} \hat{\mathbf{R}}_{xx}[k] &= x[k] x^T[k] \\ \hat{\mathbf{r}}_{x\psi}[k] &= x[k] \psi[k] \end{aligned}} \right\} \begin{array}{l} \text{Approximate expectations by} \\ \text{instantaneous samples} \end{array}$$
- Goals:
 - find an algorithm that always moves its coefficients towards the optimum solution
 - Least-Mean-Square (LMS) algorithm

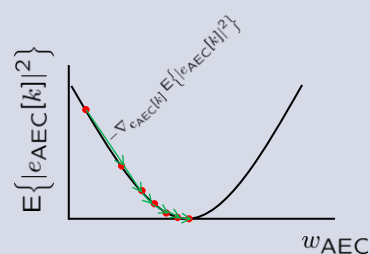
Gradient Algorithms

- LMS
- NLMS
- Many others

Gradient

$$\begin{aligned} w_{\text{AEC}}[k+1] &= w_{\text{AEC}}[k] - \mu \nabla_{w_{\text{AEC}}[k]} E\{|e_{\text{AEC}}[k]|^2\} \\ &= w_{\text{AEC}}[k] + \mu E\{e_{\text{AEC}}[k] x[k]\} \end{aligned}$$

- Negative Gradient $-\nabla_{w_{\text{AEC}}[k]} E\{|e_{\text{AEC}}[k]|^2\}$ always points towards minimum.



LMS Algorithm

- Procedure of the steepest descent:

$$\begin{aligned} \mathbf{w}_{\text{AEC}}[k+1] &= \mathbf{w}_{\text{AEC}}[k] - \mu \nabla_{\mathbf{w}_{\text{AEC}}[k]} E\{|e_{\text{AEC}}[k]|^2\} \\ &= \mathbf{w}_{\text{AEC}}[k] + \mu E\{e_{\text{AEC}}[k] \mathbf{x}[k]\} \end{aligned}$$

- For the practical application, the expectation value is measured by the instantaneous value:

$$\mathbf{w}_{\text{AEC}}[k+1] = \mathbf{w}_{\text{AEC}}[k] + \mu \mathbf{x}[k] e_{\text{AEC}}[k]$$

Procedure with smallest mean square error (LMS algorithm)

- Step size LMS: $0 \leq \mu \leq \frac{2}{\lambda_{\max}}$

NLMS Algorithm

- LMS properties:

- simplest algorithm $\rightarrow$ only L_{AEC} multiplications ($\mathbf{x} \cdot e$) 😊
- convergence depends on input signal $\rightarrow$ not appropriate for non-stationary signals like speech 😞
- appropriate step-size not easy to calculate (needs $\lambda_{\max}$) 😞
- **not** appropriate for AEC!

- NLMS

- convergence speed independent of the power of the input signals by normalization of step-size μ/σ_x^2 😊
- still computationally efficient ($\sim 2L_{\text{AEC}}$ multiplications, $\mathbf{x} \cdot e$, $\mathbf{x}^T \mathbf{x}$) 😊
- convergence is too slow at least for highly correlated input signals like speech 😞
- convergence depends on input signal correlation $\mathbf{R}_{\mathbf{x}\mathbf{x}}$ 😞

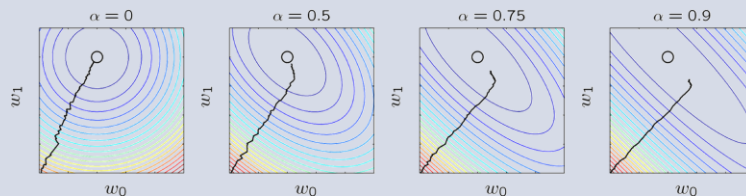
$$\mathbf{w}_{\text{AEC}}[k+1] = \mathbf{w}_{\text{AEC}}[k] + \mu \mathbf{x}[k] \frac{1}{\sigma_x^2} e_{\text{AEC}}[k]$$

$$0 \leq \mu \leq 2$$

$$\sigma_x^2 = \mathbf{x}[k]^T \mathbf{x}[k]$$

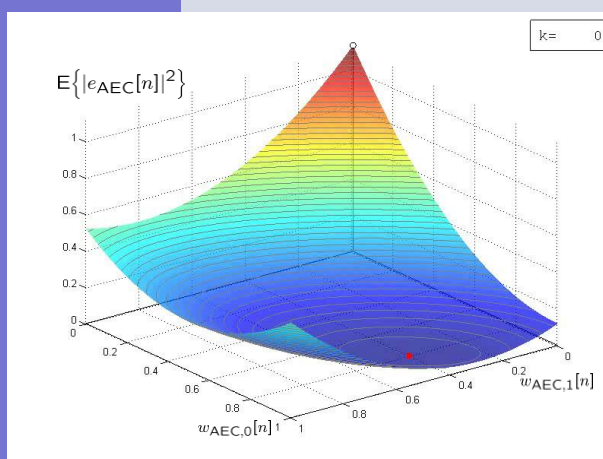
Simulation Results – LMS Algorithm's Convergence

$$\mathbf{R}_{xx} = \begin{bmatrix} 1 & \alpha \\ \alpha & 1 \end{bmatrix}$$



Traces of the filter coefficients. Convergence is worsened for a higher correlation between adjacent samples of the input signal. The algorithm was stopped after 150 iterations.

The Error Performance Surface



- Error surface for:

$$\mathbf{R}_{yy} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(last slide); Here:

$$\mathbf{R}_{yy} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

- Properties
 - Absolute minimum
 - No local minima
 - Dependence on the Autocorrelation properties of the excitation signal

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Convergence of Gradient-Algorithms (NLMS)

Convergence of AEC-Filter

ERLE in dB

time in s

Filter order 2048
Filter order 1024
Filter order 512
Filter order 128

Input: white noise, length of RIR: 2048

$\tau_{60} = 400 \text{ ms}$

Echo Return Loss Enhancement:

$$\frac{\text{ERLE}[k]}{\text{dB}} = 10\log_{10}\left(\frac{\psi[k]}{\psi[k] - \hat{\psi}[k]}\right)$$

Requirements of ITU-T G.167:

40dB suppression if near-end speaker is inactive

Maximum delay: 2ms

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Example: Convergence of an AEC

- Convergence curves: ERLE and system distance D_{dB}
- Input: white noise

Echo Return Loss Enhancement

ERLE (dB)

time in sec

System distance

D_{dB}

time in sec

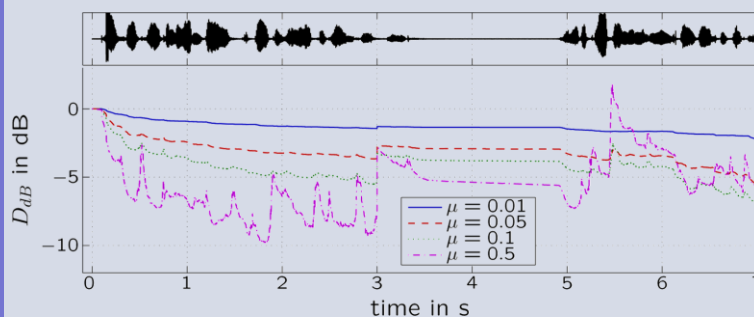
white noise
reduced white noise

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Simulation Results – NLMS Algorithm's Convergence



System distance curves for the NLMS algorithm. The step size is varied, the input signal is a speech sample.

This slide set has covered ...

- Hand-Free Systems
- Wiener Filter
- Least Mean Squares (LMS)
- Normalised LMS Algorithm (NLMS)

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Any Questions?



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Next time ...

Linear Prediction

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